

Executable Stack Models for Security Claims

Abstract

Security arguments often depend on implicit stack-state assumptions: ordering, resource ownership, transition boundaries, and failure behavior. This manuscript frames stack reasoning through executable models that expose the state space before proof obligations are stabilized.

Methodology

The model names stack states, transition rules, invariants, and failure cases in an executable form. Small traces are used to test whether each invariant is strong enough to support the claim. The formal layer then receives a smaller set of obligations with a clearer connection to the system behavior.

Results

The current record gives a compact bridge between systems behavior and formal reasoning: executable traces expose candidate invariants, and the paper records which obligations remain for proof-oriented cleanup.